

1. (25 points) Derive the Laplace transforms of the following signals from the basic integral definition (equation 3.1 of your textbook). Show your detailed derivation steps. Your answers should match those from Question 2 below.

$$\begin{aligned}x_1(t) &= e^{-3t}u(t) \\x_2(t) &= u(t - 5) \\x_3(t) &= te^{-t}u(t) \\x_4(t) &= \delta(t - 2) \\x_5(t) &= (t - 6)u(t)\end{aligned}$$

2. (25 points) Use Tables 3-1 and 3-2 as appropriate to write down the Laplace transforms of the same signals as in Question 1. Show your steps and point out the Laplace Transform properties that you have utilized for each step. Your answers should match those from Question 1.

3. (25 points) An LTI system is specified by the following differential equation:

$$\frac{d^2y(t)}{dt^2} + \frac{dy(t)}{dt} + y(t) = x(t) + \frac{dx(t)}{dt}$$

Calculate $\mathbf{H(s)} = \frac{Y(s)}{X(s)}$ for this system using Laplace Transform properties from the tables.

4. (25 points) Sketch the poles and zeros on the $\mathbf{s}$ -domain plot (the complex (σ, ω) plane) for the following function: $\mathbf{H(s)} = \frac{(s+1)(s-2)}{(s+5)(s-4)((s+1)^2+9)}$. For each pole and zero, examine its location in the $\mathbf{s}$ -plane and comment on its contribution to this system's behavior.